

AHRID

An adaptive human-risk intelligence dashboard with live threat-feed integration and behavioural risk scoring for non-technical SME staff in Kathmandu Valley, fusing OSINT threat APIs with Random Forest and clustering on user telemetry, assessed on prediction accuracy, awareness uplift, and usability, bounded by surveillance limits and risk-score transparency.

Module: ST6047CEM Cyber Security Project
Thesis Proposal

Prepared by
Rupesh Kumar Thakur
College ID: 230548
CU ID: 14806490
Batch: 35'C

Submitted To
Manoj Shrestha

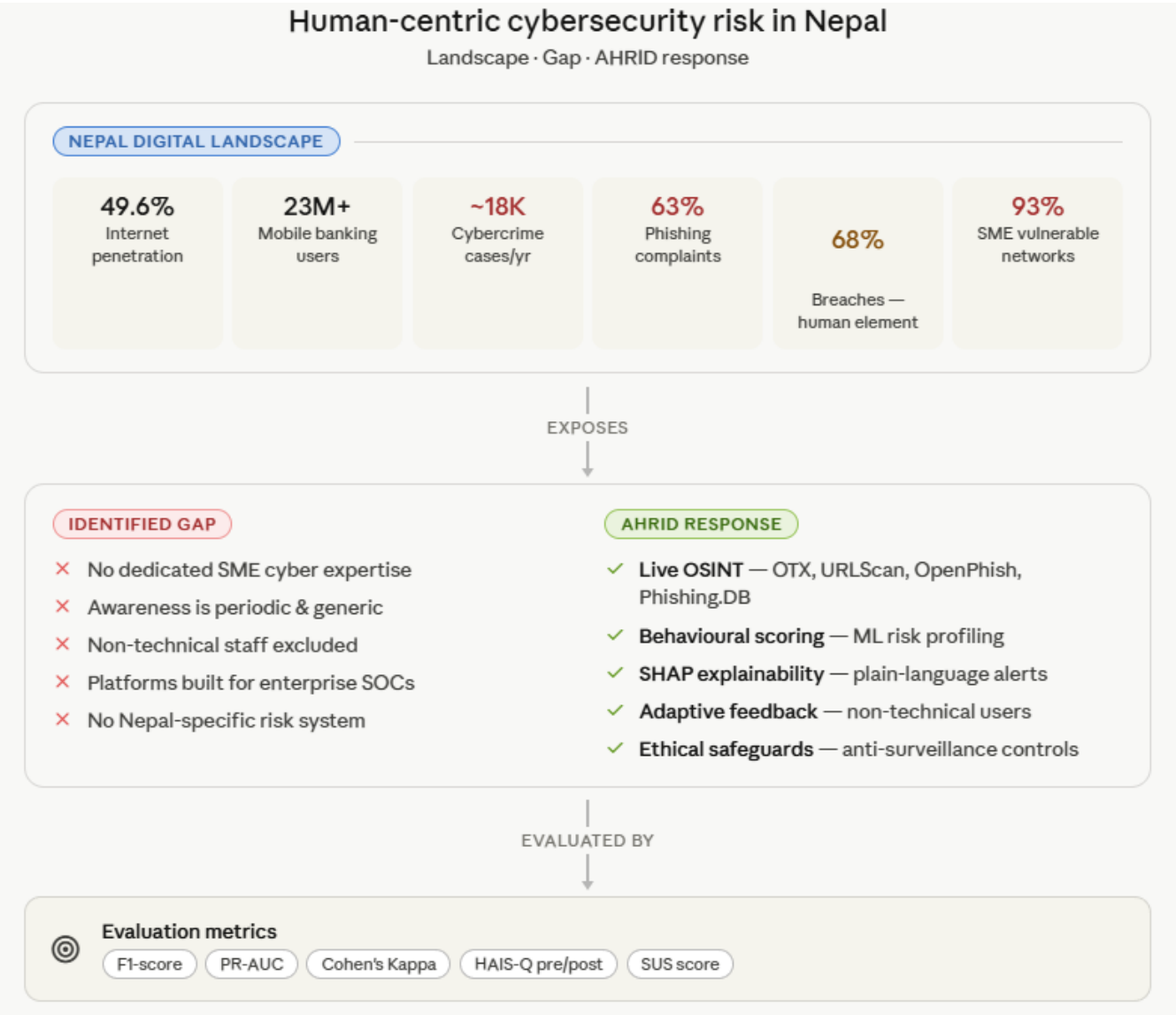


TABLE OF CONTENTS

1. Introduction.....	3
2. Background and problems statement.....	4
3. Theories & Behavioural Factors	5
4. Research Aim.....	5
5. Research Objectives.....	5
6. Justification & Significance	6
7. Research Questions.....	7
8. Hypotheses.....	7
9. Research Methodology	8
10. Data Sources & Collection Strategy	9
11. Machine Learning & Analysis Approach	10
12. Literature Review.....	12
15. Case Studies: Existing Human Risk Management Platforms	13
16. Tools & Technologies.....	14
17. Ethical Considerations	15
18. Conclusion	16
References.....	17

1. INTRODUCTION

Enterprise cybersecurity has become a critical global concern as organizations increasingly depend on cloud services, mobile banking, e-commerce platforms, and interconnected digital systems, significantly expanding the attack surface for cyber threats. At the global level, cybercrime continues to escalate alongside digital transformation, exposing structural limitations in existing security approaches.



In Nepal, digital adoption is also rapidly increasing, with internet penetration reaching 49.6% in early 2024 (DataReportal, 2024). However, this growth has been accompanied by a sharp rise in cybercrime. Recent Nepal Police Cyber Bureau based analyses report approximately 18,000–19,000 cybercrime cases annually in 2024/25, with phishing, credential theft, impersonation, and online financial fraud among the most common attacks (Dhital, 2025). These incidents increasingly exploit human behaviour through social engineering rather than technical system vulnerabilities.

Despite growing investment in cybersecurity, organisations continue to experience breaches due to reactive security models, fragmented monitoring systems, and reliance on human judgment, which limit timely detection and risk prioritisation. These challenges are more severe in small and medium enterprises (SMEs), where limited technical expertise and resources constrain effective cybersecurity management.

Existing research consistently identifies human behaviour as a key vulnerability in cybersecurity, particularly in resource-constrained environments where awareness and structured training are limited (Khadka, 2024). As a result, non-technical employees remain highly exposed to phishing and social engineering attacks.

To address this gap, this study develops and evaluates an Adaptive Human-Risk Intelligence Dashboard (AHRID) for SMEs in Kathmandu Valley. The framework integrates live threat intelligence with behavioural risk assessment to provide personalised and explainable cybersecurity risk insights for non-technical users, while also examining ethical concerns related to transparency, privacy, and workplace surveillance.

2. BACKGROUND AND PROBLEMS STATEMENT

Modern organisations, particularly small and medium enterprises (SMEs), are increasingly dependent on digital systems, making cybersecurity a critical operational concern where threats target both technical infrastructure and human behaviour. As digital adoption grows, cyber risks have shifted from isolated technical exploits to continuous, intelligence-driven attacks that exploit organisational and human vulnerabilities.

In Nepal, cybersecurity is governed by the Electronic Transactions Act 2008 and the National Cyber Security Policy 2023. However, limited institutional capacity and implementation constraints continue to affect effective cyber defence, particularly within SMEs.

SMEs in Kathmandu Valley are rapidly adopting mobile banking, cloud services, and e-commerce platforms. While this enhances efficiency, it also increases exposure to phishing, credential theft, malware, and social engineering attacks. National reports indicate a sustained rise in such incidents, with attackers increasingly targeting human decision-making rather than system weaknesses.

Despite growing risks, SMEs remain constrained by limited cybersecurity expertise, lack of dedicated security teams, and absence of real-time monitoring systems. Security practices are largely reactive and dependent on generic awareness training, which is insufficient against evolving attack strategies.

The core problem is not the absence of cybersecurity tools, but the lack of context-aware, adaptive, and user-centred systems that translate cyber risk into actionable insights for non-technical SME employees. Existing enterprise solutions are often complex, resource-intensive, and misaligned with SME operational realities in developing contexts like Nepal.

This study addresses this gap by proposing an Adaptive Human-Risk Intelligence Dashboard (AHRID) that integrates live threat intelligence with behavioural risk analysis to support real-time, explainable cybersecurity decision-making.

3. THEORIES & BEHAVIOURAL FACTORS

Cybersecurity is not purely a technical challenge, it is a form of organisational decision-making under uncertainty and adversarial pressure. Non-technical SME employees often face cognitive limitations when responding to evolving threats, which can result in suboptimal risk decisions. The Adaptive Human-Risk Intelligence Dashboard (AHRID) is grounded in behavioural theories that explain why individuals misinterpret or ignore cyber risks and how systems can support better decision-making.

Protection Motivation Theory (PMT) explains how threat appraisal and coping appraisal influence protective behaviour. Employees are more likely to act when threats are perceived as relevant and protective measures as feasible. AHRID translates live threat intelligence and behavioural risk indicators into simplified, actionable guidance, addressing situations where employees often struggle to interpret conventional security alerts.

Prospect Theory demonstrates that individuals' overweight potential losses relative to equivalent gains. SME employees are more responsive to warnings framed as personal or organisational losses — e.g., account compromise or financial exposure — than to abstract benefits. AHRID employs loss-framed messaging strategies to improve engagement while avoiding excessive fear that could reduce long-term compliance.

Technology Acceptance Model (TAM) highlights perceived usefulness and ease of use as key determinants of technology adoption. AHRID incorporates these principles to ensure usability and acceptance among non-technical staff, balancing operational effectiveness with simplicity and transparency.

Together, these theories form the behavioural foundation of the AHRID design framework. The system augments human decision-making rather than replacing it, presenting personalised and explainable cybersecurity risk insights that empower employees while reducing errors caused by cognitive biases and misaligned threat awareness.

4. RESEARCH AIM

The aim of this research is to design, develop, and evaluate an Adaptive Human-Risk Intelligence Dashboard (AHRID) that integrates live OSINT threat feeds with Random Forest classification and behavioural clustering of user telemetry to provide real-time, personalised, and explainable cybersecurity risk insights for non-technical SME staff in Kathmandu Valley, while assessing system performance through prediction accuracy, awareness uplift, and usability, and critically examining ethical boundaries including employee surveillance, transparency, and responsible deployment of behavioural risk scoring.

5. RESEARCH OBJECTIVES

The following objectives have been formulated to achieve the research aim. Each objective is categorised into **Learn & Understand**, **Design & Develop**, or **Feedback & Submit**, mapping clearly to the relevant chapters and research questions.

#	Objective	Type	Maps To
1	Analyse the cybersecurity threat landscape for non-technical SME staff in Kathmandu Valley, focusing on social engineering threats, organisational security weaknesses, and behavioural risk factors.	Learn & Understand	Literature Review
2	Investigate publicly available OSINT threat intelligence sources including openphish, urlscan, AlienVault OTX, and Phishing.Database and evaluate their suitability for real-time integration into SME-focused cybersecurity systems.	Learn & Understand	Literature Review
3	Examine existing human-risk management and security awareness platforms such as CybSafe, KnowBe4, and Hoxhunt to identify limitations in explainability, accessibility, local threat adaptation, and behavioural risk communication.	Learn & Understand	Case Studies
4	Design, develop, and evaluate the Adaptive Human-Risk Intelligence Dashboard (AHRID) by integrating live OSINT feeds with Random Forest classification, behavioural clustering, and explainable risk-scoring, and assess performance through prediction accuracy, awareness uplift, and System Usability Scale (SUS) metrics.	Design & Develop	RQ1 · H1
5	Investigate the ethical implications of deploying behavioural risk scoring systems in SME environments, focusing on employee surveillance, transparency, and responsible deployment, and document findings, limitations, and governance considerations in the final thesis.	Feedback & Submit	RQ2 · H2

6. JUSTIFICATION & SIGNIFICANCE

Firstly, Nepal's cybersecurity deficit is concentrated in the human layer of SMEs. Cybercrime has grown over 300% in five years, with 93% of vulnerable networks concentrated in SMEs. Research targeting non-technical employee risk behaviour addresses a critical, underexplored vulnerability in Nepal's SME cybersecurity landscape. The AHRID framework is tailored to this organisational and geographic context, providing a contribution that generic academic ML research cannot substitute.

Secondly, integrating live OSINT threat intelligence with behavioural machine learning in a unified dashboard represents a significant engineering challenge. OSINT API integration introduces rate-limiting, data freshness, and cross-source consistency issues. Applying Random Forest classification and k-means clustering to behavioural telemetry introduces class imbalance, high-dimensional feature engineering, and interpretability requirements that conventional datasets cannot simulate. This technical complexity is both appropriate and achievable within a BSc final-year project using public data and open-source tools.

Thirdly, the ethical implications of behavioural risk scoring particularly the risk of employee surveillance remain underexplored. By investigating transparency, accountability, and responsible deployment, this research contributes to emerging frameworks in responsible AI for SME cybersecurity.

Fourthly, the project is feasible. All OSINT feeds are publicly accessible, synthetic user telemetry will simulate SME scenarios, and awareness and usability testing will involve volunteer participants with pseudonymised data under institutional ethical approval.

Finally, this research develops career-relevant expertise in ML-driven security analytics, OSINT integration, and responsible AI deployment skills aligned with the projected 13% annual growth of Nepal's cybersecurity market through 2029. Additionally, it bridges computer science, behavioural psychology, and organisational risk management, contributing interdisciplinary insights beyond technical implementation.

7. RESEARCH QUESTIONS

RQ1 (Technical)

To what extent can an Adaptive Human-Risk Intelligence Dashboard (AHRID), integrating live OSINT threat-feed data with Random Forest classification and k-means behavioural clustering on SME-representative user telemetry, produce risk scores that are statistically accurate in classifying user risk tiers, measurable improvements in cybersecurity awareness and behavioural intention, and usability levels sufficient to support voluntary adoption in resource-constrained SME environments?

RQ2 (Ethical)

What governance frameworks and technical design constraints are required to ensure that automated behavioural risk scoring in SME environments does not constitute illegitimate workplace surveillance, and how can transparency mechanisms be designed to satisfy both the informational needs of non-technical employees and the accountability requirements of SME owners under Nepalese regulations?

8. HYPOTHESES

H1 (Technical)

A Random Forest classifier trained on OSINT-enriched user telemetry will achieve risk-tier classification accuracy exceeding traditional rule-based approaches by at least 15 percentage points (F1-score, Precision-Recall AUC on a held-out test set), and interaction with the AHRID dashboard will produce a statistically significant improvement in cybersecurity awareness scores among non-technical SME participants (paired t-test, $p < 0.05$, minimum effect size Cohen's $d = 0.5$), while maintaining a System Usability Scale score of at least 68.

H2 (Ethical)

Automated behavioural risk scoring systems in SME environments introduce measurable surveillance and risk-score opacity harms that cannot be fully mitigated by technical transparency alone. Complementary governance mechanisms comprising consent protocols, score-access boundaries, and

appeal procedures are required, and their adequacy can be empirically evaluated via structured stakeholder feedback from SME owners and non-technical employees.

9. RESEARCH METHODOLOGY

This research adopts a **desk-based agile methodology**. The desk-based nature reflects that all threat intelligence data is sourced from publicly accessible OSINT APIs and academic datasets, and no primary collection from live enterprise networks is required. User evaluation is conducted via structured assessments with volunteer SME participants, rather than passive monitoring, which simplifies the ethical approval process while maintaining research rigour.

The **agile approach**, adapted from software engineering, structures development into **two-week sprints**. Each sprint produces a testable increment of the Adaptive Human-Risk Intelligence Dashboard (AHRID), with sprint reviews enabling course correction if API integration, feature engineering, or usability findings challenge initial assumptions. This iterative model is especially suited to machine learning development, where initial design choices often require adjustment when applied to real-world OSINT-enriched user telemetry.

Evaluation Design

Evaluation follows a comparative mixed-methods design, linking directly to the research questions and hypotheses:

Technical Performance (RQ1/H1): Random Forest classifier performance is compared against a rule-based scoring baseline using F1-score, Precision-Recall AUC, and Cohen’s Kappa.

Awareness Uplift (RQ1/H1): Measured via validated pre-test/post-test instruments administered to SME participants.

Usability (RQ1/H1): Assessed through the System Usability Scale (SUS).

Ethical Risk (RQ2/H2): Semi-structured interviews with SME owners (n = 5) and thematic analysis of open-ended participant responses.

Sprint Plan

Sprint	Activities
1–2	Literature review, OSINT API scoping, dataset acquisition
3–4	Synthetic user telemetry generation, feature engineering, class balancing
5–6	Random Forest model development and cross-validation
7–8	K-means clustering for behavioural persona identification

9	OSINT API live integration and data pipeline
10	SHAP explainability layer and dashboard front-end development
11–12	User evaluation (awareness uplift + SUS), SME owner interviews
13	Ethical analysis and governance framework development
14–15	Thesis writing, review, submission

10. DATA SOURCES & COLLECTION STRATEGY

The AHRID framework will be developed and evaluated using a combination of publicly accessible threat intelligence feeds, synthetic behavioural telemetry, threat pattern datasets, and structured human evaluation data. This multi-source approach ensures that each component of the system contributes a distinct signal to the integrated risk intelligence pipeline, supporting both technical performance evaluation and ethical analysis.

Primary Data Sources

Live OSINT Threat Intelligence Feeds

1. AlienVault OTX: Supplies pulse-based threat indicators mapped to MITRE ATT&CK TTPs and geolocation-enriched threat actor data, enabling community-driven detection of emerging threats across IPs, URLs, domains, and file hashes.
2. URLScan: Provides detailed behavioural analysis of URLs and domains by actively scanning and rendering submitted links, delivering confidence-scored assessments of malicious or suspicious web content.
3. OpenPhish: A continuously updated phishing intelligence feed that identifies and tracks active phishing URLs in real time, particularly effective for detecting credential-harvesting campaigns and brand-targeted attacks.
4. Phishing.Database: An open-source, community-maintained repository of confirmed phishing domains and URLs, offering broad coverage for filtering known malicious indicators and enhancing signal quality for risk scoring.

Synthetic User Behavioural Telemetry

Generated programmatically from documented SME security incident profiles and the CERT Insider Threat Dataset. Features include login time distributions, URL click patterns, email interaction rates, credential change frequency, and device access anomalies.

Synthetic generation avoids ethical and practical barriers associated with monitoring live employees, while SMOTE (Synthetic Minority Over-sampling Technique) addresses extreme class imbalance between high- and low-risk behaviours.

Supplementary Data Sources

Threat pattern datasets

Openphish: provides labelled phishing urls for supervised classification training.

Mitre att&ck enterprise matrix: serves as a structured mapping layer, linking detected behavioural anomalies to documented adversary tactics, techniques, and procedures (TTPS), ensuring operational relevance of AHRID outputs.

Human evaluation data

Awareness and usability: structured Pre-Test/Post-Test instrument based on the Hais-Q questionnaire administered to ~30 volunteer sme employees. System usability scale (sus) administered immediately post-interaction with the dashboard.

Ethical governance assessment: semi-structured interviews with five SME owners provide qualitative insight into transparency, consent, and operational governance requirements.

11. MACHINE LEARNING & ANALYSIS APPROACH

The AHRID framework uses a dual-component ML architecture to classify individual employee risk tiers and uncover latent behavioural risk patterns.

Supervised Component: Random Forest Classification

A Random Forest classifier is trained on OSINT-enriched telemetry to assign low, medium, or high risk tiers. Random Forest is chosen for high-dimensional, class-imbalanced datasets and feature importance ranking for SHAP explainability. Hyperparameters are tuned via stratified 10-fold cross-validation. Performance is evaluated using F1-score, Precision-Recall AUC, and Cohen's Kappa.

Unsupervised Component: K-Means Clustering

K-means identifies latent behavioural personas from telemetry data. Optimal cluster count k is selected via elbow and silhouette methods. Cluster membership is added as a feature to the Random Forest, forming a hybrid supervised-unsupervised pipeline that improves predictive performance and enables persona-specific risk communication.

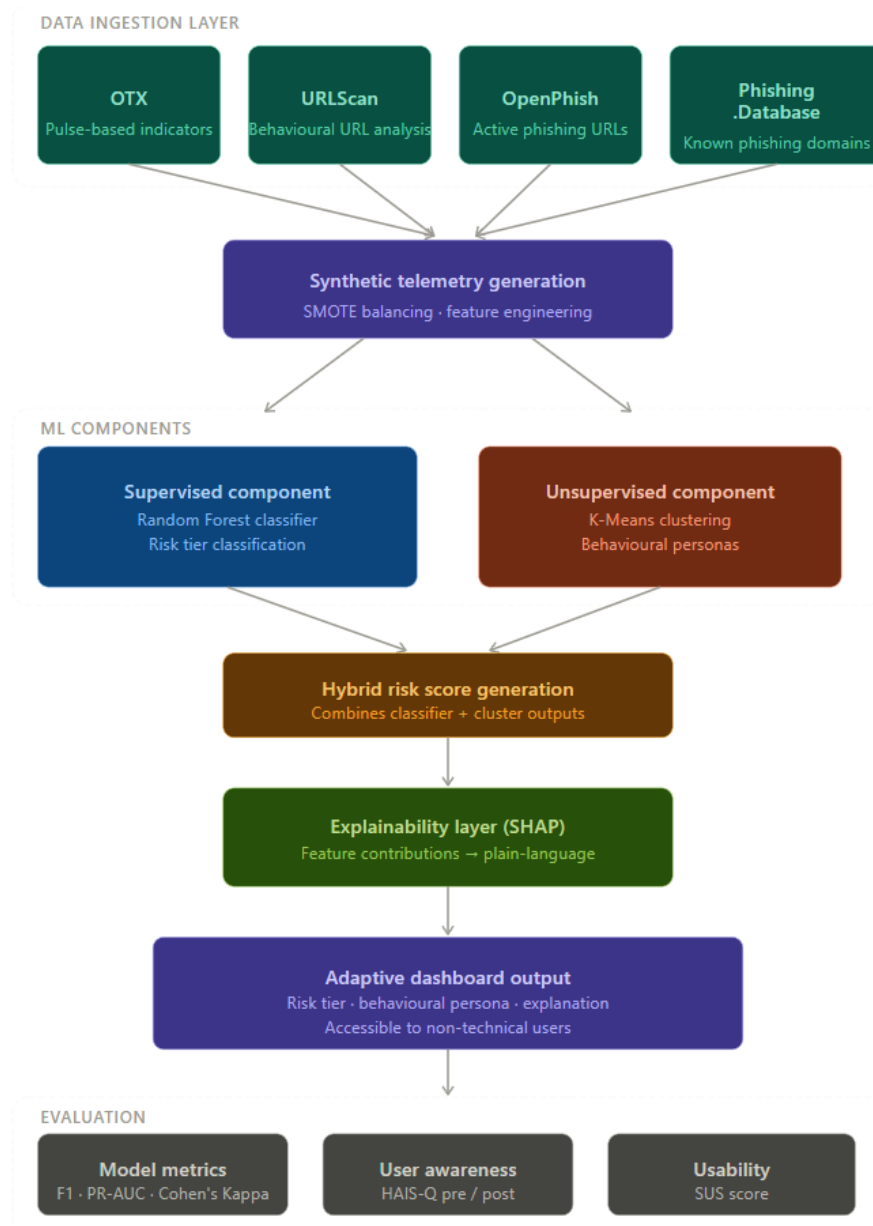
Explainability Layer: SHAP

SHAP values provide per-feature contribution for each risk score. The dashboard translates these into plain-language explanations, ensuring transparency and user understanding, directly addressing RQ2.

Class Imbalance Handling

SMOTE is applied to synthetic telemetry to mitigate rare high-risk behaviour underrepresentation. Metrics are chosen for robustness under class imbalance.

AHRID Architecture



12. LITERATURE REVIEW

LITERATURE REVIEW: HUMAN RISK IN SME CYBERSECURITY

Non-technical employees in SMEs represent both the primary attack vector and the least-targeted beneficiary of cybersecurity investment (Parsons et al., 2017). The HAIS-Q provides a psychometrically validated instrument for assessing security awareness, attitudes, and behaviour across seven domains (passwords, email, social networking, incident reporting), directly informing the pre/post evaluation framework for the AHRID dashboard.

Cognitive failures attentional lapses, memory slips, and inhibition errors predict risky cybersecurity behaviour independently of knowledge (Hadlington, 2017). This highlights the design imperative for risk communication that reduces cognitive load, provides actionable guidance, and avoids abstract threat descriptions.

SMEs in developing economies face structural barriers: lack of dedicated security personnel, inability to interpret commercial security tools, and non-contextual, static training programmes (Kabanda et al., 2018 & Rombaldo Junior et al., 2023). Nepal-specific studies show SMEs rely on outdated technology, lack formal policies, and provide minimal training (Adhikari et al., 2024). These findings justify AHRID's focus on delivering security intelligence accessible to non-technical users, enabling immediate, context-relevant action.

LITERATURE REVIEW: MACHINE LEARNING IN BEHAVIOURAL RISK SCORING

Random Forests, foundationally described by Breiman (2001), perform robustly on high-dimensional, class-imbalanced datasets matching the characteristics of OSINT-enriched user telemetry. Ensemble methods outperform single classifiers in cybersecurity datasets (Buczak & Guven, 2016), supporting AHRID's supervised component.

Clustering algorithms reveal latent behavioural personas. K-means, while less sophisticated than density-based methods such as HDBSCAN, offers superior interpretability for non-technical stakeholders, aligning with the dashboard's target audience (Artioli et al., 2024). Cluster membership augments supervised classification, forming a hybrid pipeline that improves predictive performance while maintaining transparency.

SHAP values provide per-prediction feature attribution, satisfying consistency and local accuracy axioms (Lundberg & Lee, 2017). This enables user-specific explanations, critical for ethical transparency and actionable risk communication.

A key limitation is reliance on enterprise-scale datasets: behavioural telemetry in SMEs is smaller, less stable, and subject to distribution shifts and concept drift (Sommer & Paxson, 2010). AHRID mitigates this by combining synthetic telemetry with SME-specific evaluation, acknowledging operational risk within the governance framework.

LITERATURE REVIEW: BEHAVIOURAL ECONOMICS & SECURITY DECISION-MAKING

Protection Motivation Theory (Rogers, 1975, 1983) explains cybersecurity protective behaviour, emphasising that coping self-efficacy often outweighs threat awareness (Workman et al., 2008). Risk scores without actionable guidance may increase threat perception but reduce coping appraisal, risking learned helplessness.

Prospect Theory (Kahneman & Tversky, 1979), applied to security compliance through the “compliance budget” concept (Beautement et al., 2008), explains routine insecure workarounds: employees systematically overweight effort costs relative to breach likelihood. AHRID presents contextualized OSINT-derived breach probabilities to improve threat appraisal accuracy and influence compliance behaviour positively.

Perceived ease of use predicts adoption (Davis, 1989 & Venkatesh & Bala, 2008). Usability is therefore a core research criterion: a technically accurate dashboard scoring below a SUS of 68 would fail as a security intervention, regardless of ML performance.

15. CASE STUDIES: EXISTING HUMAN RISK MANAGEMENT PLATFORMS

CybSafe (Behavioural Risk Scoring and Departmental Analytics)

CybSafe’s Security Behaviour Database tracks over 100 security behaviours and generates composite human cyber risk scores disaggregated by department, role, and peer group. This validates AHRID’s user-level behavioural clustering approach. However, CybSafe does not integrate live threat intelligence: risk scores reflect observed behaviour and simulated phishing results, independent of evolving external threats. AHRID addresses this gap by fusing live OSINT signals with behavioural telemetry, ensuring risk scores reflect both user behaviour and current threat conditions.

KnowBe4 (Phishing Simulation and Security Culture Measurement)

KnowBe4 provides global-scale phishing simulations and Security Culture Surveys, with PhishER automating triage of user-reported phishing emails. This demonstrates the commercial viability of behavioural risk scoring for non-technical employees. Limitations include proprietary, opaque scoring algorithms and the absence of per-user explainability. These gaps directly motivate AHRID’s SHAP-based explainability layer, ensuring transparency and ethical deployment in regulated or privacy-sensitive SME environments.

Hoxhunt (Adaptive, Behaviour-Responsive Micro training)

Hoxhunt’s adaptive micro-training platform demonstrates that personalised, behaviour-responsive interventions outperform generic periodic training. Forrester Wave 2024 recognition confirms the commercial maturity of behaviour-adaptive awareness approaches. However, Hoxhunt requires enterprise email infrastructure and SIEM integration, creating adoption barriers for SMEs. None of the examined platforms are designed for, evaluated in, or adapted to the Kathmandu Valley SME context a primary target for AHRID deployment.

16. TOOLS & TECHNOLOGIES

Tools are selected according to four explicit criteria: open-source availability under academic-compatible licences, documented effectiveness for cybersecurity or behavioural ML tasks, compatibility with OSINT APIs, and support for explainability requirements in ethical analysis.

Category	Tool / Framework	Purpose
Programming & ML Framework	Python 3.11	Serves as the primary development language, offering dominant support in ML and API integration.
	Scikit-learn	Provides Random Forest classification, k-means clustering, SMOTE handling, and cross-validation capabilities.
	SHAP	Enables post-hoc explainability with per-user feature attribution, supporting ethical transparency in risk communication.
	Pandas / NumPy	Used for feature engineering, data preprocessing, and synthetic behavioural telemetry generation.
OSINT API Integration	AlienVault OTX	Supplies pulse-based TTP indicators with MITRE ATT&CK mapping and geolocation-enriched threat actor data.
	URLScan	Provides reputation scoring and behavioural analysis for URLs and domains by actively scanning and rendering submitted links.
	OpenPhish	Delivers real-time phishing URL intelligence, distinguishing active credential-harvesting campaigns from known historical indicators.
	Phishing.Database	An open-source community-maintained repository supplying confirmed phishing domains and URLs for broad indicator coverage and confidence scoring.
Dashboard & Deployment	Flask + React	Implements a lightweight web stack suitable for low-bandwidth SME environments.
	SQLite	Offers local risk score storage without reliance on enterprise database infrastructure.
	Docker	Facilitates containerised deployment to ensure reproducibility and environmental consistency.
Evaluation & Baseline Instruments	HAIS-Q	Provides pre/post security awareness measurement.
	System Usability Scale (SUS)	Conducts standardised 10-item usability assessment.
	Optional Baseline Tools	Includes lightweight network analysis or synthetic telemetry validation tools (e.g., Scapy, custom log generators) to support reproducibility and comparative evaluation.

17. ETHICAL CONSIDERATIONS

Overview

This research addresses ethical considerations across two dimensions: research process ethics, governing data collection and use, and deployment ethics, governing AHRID's operational implications in SME environments.

Research Process Ethics

All OSINT threat intelligence is sourced from publicly accessible APIs under terms permitting academic research. No personal data from identifiable individuals is collected without informed consent. Synthetic user telemetry is generated from published SME incident profiles, requiring no live network access. Human participants are volunteer SME employees, providing informed consent, retaining the right to withdraw, and contributing anonymised data. Data is retained only for thesis analysis and deleted upon completion.

Deployment Ethics

Deployment ethics are addressed in four dimensions:

Privacy and Surveillance Boundaries: Individual behavioural scores are sensitive only aggregate anonymised scores are used for managerial reporting. Scores must not influence employment or access decisions without explicit governance. Employees are informed of all telemetry sources. Nepal's Individual Privacy Act 2075 (2019) provides the legislative baseline.

Transparency: SHAP explainability provides per-user feature attribution. The research evaluates whether explanations are comprehensible to non-technical employees, with redesign and re-evaluation conducted where comprehension gaps exist.

Algorithmic Fairness: Historical behaviour patterns may disadvantage certain employees, such as shift workers or those with accessibility requirements. Risk scores are analysed across subgroups to identify and document disparities.

Human-AI Collaboration: AHRID operates as a human-in-the-loop advisory system. All consequential decisions remain with human managers. Ethical risks of automation, including false positives and negatives, are documented and incorporated into the governance framework.

18. CONCLUSION

This proposal has outlined a research project to design, develop, and evaluate the Adaptive Human-Risk Intelligence Dashboard (AHRID) for non-technical SME staff in the Kathmandu Valley. The primary contribution of this research is not a commercially deployable security product, but a rigorous, empirically grounded understanding of how behavioural clustering, Random Forest classification, and live OSINT threat intelligence can be integrated to improve cybersecurity risk awareness and behavioural intention among non-technical employees.

The project treats technical, behavioural, and ethical dimensions as co-equal investigative threads. The governance framework developed in response to RQ2 informs AHRID's architecture from the outset the usability evaluation guided by Protection Motivation Theory shapes interface design throughout development and empirical measurement of awareness uplift provides evidence of real-world impact beyond technical metrics.

Grounded in the security challenges specific to Kathmandu's SME sector, this research applies robust theoretical frameworks from Protection Motivation Theory, Prospect Theory, and the Technology Acceptance Model. Ethical considerations including privacy, fairness, transparency, and human-in-the-loop decision-making are integrated into every stage, ensuring that AHRID respects surveillance boundaries and operational transparency.

In summary, this project offers an opportunity to apply interdisciplinary knowledge from cybersecurity, machine learning, and behavioural science to a pressing practical problem, while developing critical thinking, ethical reasoning, and implementation skills expected of a distinction-level BSc contribution. The emphasis is on learning and understanding the complexity of SME cybersecurity, rather than delivering a definitive commercial solution, reflecting the intellectual honesty and scholarly rigor of the research.

REFERENCES

- Adhikari, S.N. and Molla, N. (2024) 'Cybersecurity challenges in small and medium enterprises (SMEs) in Nepal', *International Journal of Multidisciplinary Innovative Research*, 2(6). Available at: <https://www.ijmir.com/v2i6/5.php> (Accessed: 9 May 2026).
- Artioli, P., Maci, A. and Magri, A. (2024) 'A comprehensive investigation of clustering algorithms for user and entity behaviour analytics', *Frontiers in Big Data*, 7, article 1375818. doi: 10.3389/fdata.2024.1375818.
- Beautement, A., Sasse, M.A. and Wonham, M. (2008) 'The compliance budget: managing security behaviour in organisations', *Proceedings of the 2008 Workshop on New Security Paradigms*, pp. 47–58.
- Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32. doi: 10.1023/A:1010933404324.
- Brooke, J. (1996) 'SUS: A quick and dirty usability scale', in Jordan, P.W. et al. (eds.) *Usability Evaluation in Industry*. London: Taylor & Francis, pp. 189–194.
- Buczak, A.L. and Guven, E. (2016) 'A survey of data mining and machine learning methods for cyber security intrusion detection', *IEEE Communications Surveys & Tutorials*, 18(2), pp. 1153–1176. doi: 10.1109/COMST.2015.2494502.
- Davis, F.D. (1989) 'Perceived usefulness, perceived ease of use, and user acceptance of information technology', *MIS Quarterly*, 13(3), pp. 319–340. doi: 10.2307/249008.
- Hadlington, L. (2017) 'Human factors in cybersecurity examining the link between Internet addiction, impulsivity, attitudes towards cybersecurity, and risky cybersecurity behaviours', *Heliyon*, 3(7), e00346. doi: 10.1016/j.heliyon.2017.e00346.
- Kabanda, S., Tanner, M. and Kent, C. (2018) 'Exploring SME cybersecurity practices in developing countries', *Journal of Organizational Computing and Electronic Commerce*, 28(3), pp. 266–282. doi: 10.1080/10919392.2018.1480380.
- Kahneman, D. and Tversky, A. (1979) 'Prospect Theory: An analysis of decision under risk', *Econometrica*, 47(2), pp. 263–291. doi: 10.2307/1914185.
- Lundberg, S.M. and Lee, S.I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30, pp. 4765–4774.
- Ministry of Communication and Information Technology (2023) *National Cyber Security Policy 2080*. Kathmandu: Government of Nepal. Available at: <https://mocit.gov.np> (Accessed: 9 May 2026).
- Parsons, K., McCormac, A., Butavicius, M., Pattinson, M. and Jerram, C. (2017) 'The development of the Human Aspects of Information Security Questionnaire (HAIS-Q): A pilot study', *Computers & Security*, 64, pp. 246–257. doi: 10.1016/j.cose.2016.09.012.
- Rogers, R.W. (1975) 'A protection motivation theory of fear appeals and attitude change', *Journal of Psychology*, 91(1), pp. 93–114. doi: 10.1080/00223980.1975.9915803.
- Rombaldo Junior, C., Becker, I. and Johnson, S. (2023) 'Unaware, unfunded and uneducated: A systematic review of SME cybersecurity', *arXiv:2309.17186 [cs.CR]*. Available at: <https://arxiv.org/abs/2309.17186> (Accessed: 9 May 2026).
- Sommer, R. and Paxson, V. (2010) 'Outside the closed world: On using machine learning for network intrusion detection', *Proceedings of the 2010 IEEE Symposium on Security and Privacy*, pp. 305–316. doi: 10.1109/SP.2010.25.

Verizon (2024) 2024 Data Breach Investigations Report. New York: Verizon Communications. Available at: <https://www.verizon.com/business/resources/reports/dbir/> (Accessed: 9 May 2026).

Venkatesh, V. and Bala, H. (2008) 'Technology Acceptance Model 3 and a research agenda on interventions', *Decision Sciences*, 39(2), pp. 273–315. doi: 10.1111/j.1540-5915.2008.00192.x.

Workman, M., Bommer, W.H. and Straub, D. (2008) 'Security lapses and the omission of information security measures: A threat control model and empirical test', *Computers in Human Behavior*, 24(6), pp. 2799–2817. doi: 10.1016/j.chb.2008.04.006.